

Secure network slicing for distribution grid monitoring in energy utilities, ensuring isolation and resilience against cyber threats to meet NERC CIP compliance.

Energy × Critical infrastructure monitoring × Network slicing · OS081 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
62 is the world moving	46 can we play, can we win	MEDIUM 0 named in the evidence	€171m partial evidence · per year	LATER research and standards activi...	L1 13 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Evidence gap (SC-13). Orange has 3.0 published reference(s) in Energy, below the threshold of 5. A customer conversation here has no proof point behind it yet.

The opportunity

Secure network slicing for distribution grid monitoring in energy utilities, ensuring isolation and resilience against cyber threats to meet NERC CIP compliance. This opportunity exists now because research and standards activity is actively addressing the security of 5G network slicing for smart grids, yet the market is still early and standards are not fully mature.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€207m €32.7m – €3.2bn	€171m €27.0m – €2.6bn	€2.1m €324k – €31.7m	partial
Observed: tendered contract value, annualised	€172m €172m – €460m	€172m €172m – €460m	€2.1m €2.1m – €5.5m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Energy (EU27_2020)	5,796 enterprises	observed	Eurostat · sbs_sc_oww	2024
Enterprises adopting Network slicing	51.7 % of enterprises (10+ employees)	proxy	Eurostat · isoc_eb_iotn2 · E_IOT1	2021
Annual value of one engagement	€299k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Size-weighted buyer base	1,341 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Internet of Things by NACE Rev. 2 activity series E_IOT1 is used as a proxy for Network slicing: No slicing series; industrial IoT use as proxy. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

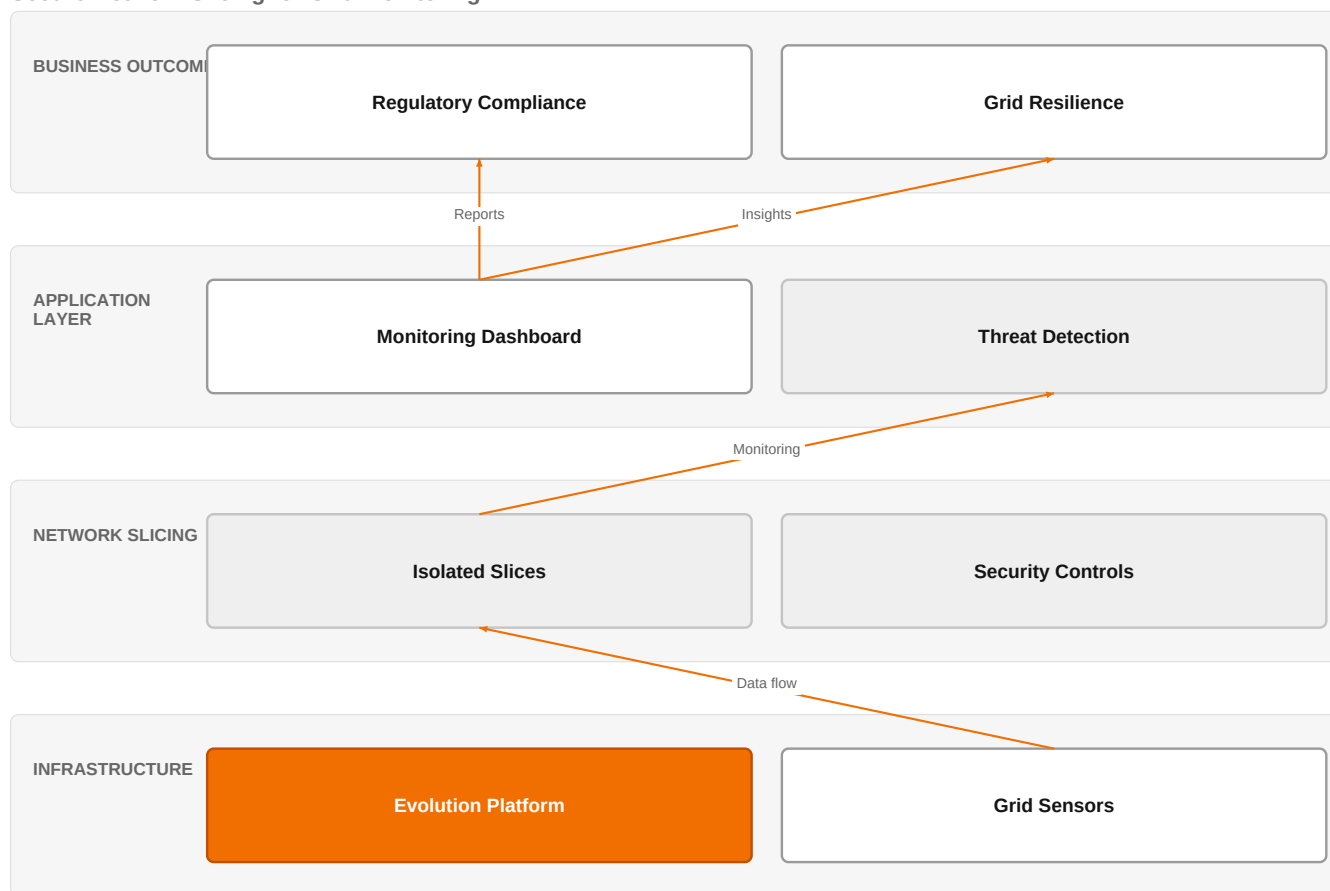
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€172m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 21 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

The solution, in one picture

Secure Network Slicing for Grid Monitoring



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This diagram shows how Orange's secure network slicing isolates grid monitoring traffic, enabling compliance and resilience.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Research and standards are actively addressing the security of 5G network slicing for smart grid networks, as evidenced by recent publications. The increasing adoption of IIoT devices in critical infrastructure introduces significant cybersecurity and privacy challenges, particularly for anomaly detection and secure data sharing. The RIGOROUS approach proposes end-to-end network slicing for enhanced cybersecurity in 5G/6G environments, indicating a shift toward more secure slicing architectures. However, the technology is still in research and standards activity, with no mature commercial deployments yet.

Sources: SIG-D8CAA774FDC4 Next Research 2026-07-01; SIG-F633044AF87A openalex.org 2026-01-29; SIG-A5AB1DD30A6D openalex.org 2025-11-10

Who buys, and why now

The buyer is typically the Chief Information Security Officer (CISO) or the head of OT/ICS security at a large energy utility. The pain is the need to monitor distribution grids in real-time while ensuring isolation and resilience against cyber threats to meet NERC CIP compliance. The trigger is often a regulatory audit or a near-miss cyber incident that exposes vulnerabilities in the current monitoring infrastructure.

Why it is hot now — every claim bound to a dated source

- Securing 5G network slicing for the smart grid network is a key concern. [SIG-D8CAA774FDC4]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a secure network slicing solution leveraging its Cybersecurity experts, IoT experts, and Digital experts. The engagement would start with a security assessment of the utility's current grid monitoring setup, followed by the design of a network slicing architecture that isolates critical monitoring traffic. Orange would then implement the solution on its Evolution Platform, ensuring compliance with ISO 27001 and SecNumCloud certifications. The solution would include continuous monitoring and threat detection, potentially using Orange Drone Guardian for physical infrastructure inspection.

Why Orange can win

Orange can win because it combines cybersecurity expertise with IoT and digital capabilities, backed by ISO 27001 and SecNumCloud certifications. The Evolution Platform provides a secure foundation for network slicing, and Orange Drone Guardian adds a unique physical security dimension. However, the assets are relatively thin for this specific use case, and Orange lacks a dedicated energy vertical offering, so the win must be based on the strength of its security credentials and the ability to integrate multiple capabilities.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Evolution Platform	offer	L1	offer provides the technology in a matching domain	unconfirmed
Orange Drone Guardian	offer	L1	offer addresses the use case but not this technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
SecNumCloud	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
Cybersecurity experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Digital experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 6.6; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitors include telecom operators like Colt Technology Services, Deutsche Telekom / T-Systems, and Vodafone Business, who offer network slicing as part of their connectivity solutions. Network equipment vendors such as Boldyn Networks, Ericsson, Huawei, and Nokia also sell network slicing, with Nokia being active in the energy sector. Global systems integrator Capgemini is also active in energy. The customer will compare Orange against these players based on security expertise, existing relationships, and the ability to deliver end-to-end solutions.

Sources: SIG-D8CAA774FDC4 Next Research 2026-07-01

Competitor	Category	Position here	Basis	Evidence
Colt Technology Services	Telecom operator peer	sells Network slicing; competes in Connectivity Solutions	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	sells Network slicing; competes in Connectivity Solutions, Cybersecurity	structural	—
Vodafone Business	Telecom operator peer	sells Network slicing; competes in Connectivity Solutions	structural	—
Boldyn Networks	Network equipment vendor	sells Network slicing; competes in Connectivity Solutions	structural	—

Ericsson	Network equipment vendor	sells Network slicing; competes in Connectivity Solutions	structural	—
Huawei	Network equipment vendor	sells Network slicing; competes in Connectivity Solutions — also an Orange partner	structural	—
Nokia	Network equipment vendor	sells Network slicing; active in Energy; competes in Connectivity Solutions	structural	—
Capgemini	Global systems integrator	active in Energy	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 How are you currently monitoring your distribution grid, and what are the main challenges with the existing solution?
- 2 Have you experienced any cyber incidents or near-misses that have prompted a review of your monitoring infrastructure?
- 3 What is your current level of compliance with NERC CIP, and are there any upcoming audits or regulatory changes that are driving urgency?
- 4 Do you have a dedicated OT/ICS security team, and how do they currently handle threat detection and response?
- 5 What is your timeline for upgrading or transforming your grid monitoring capabilities, and what budget is allocated for this?

Objections you will meet

They will say	You can answer
Network slicing is still an emerging technology; we are not confident it is mature enough for critical infrastructure.	You are right that network slicing is still evolving, but the security of 5G network slicing for smart grids is a key concern being addressed by research. Orange can help you pilot the technology in a controlled environment to validate its reliability before full deployment.
We already have a secure monitoring solution; why would we change?	Your current solution may be secure, but network slicing offers the ability to isolate critical monitoring traffic, reducing the attack surface and improving resilience. Orange can help you assess whether your current solution meets future regulatory and threat requirements.
We are concerned about the cost of implementing a new technology.	The initial investment may be significant, but the long-term benefits of improved security and compliance can outweigh the costs. Orange can work with you to design a solution that fits your budget and provides a clear return on investment.

Risks and unknowns

The main risk is that the technology is still in research and standards activity, with no mature commercial deployments, which could stall adoption. Additionally, the regulatory landscape for NERC CIP compliance is evolving, and it is unclear how network slicing will be treated. There is also uncertainty about the customer's willingness to adopt a relatively new technology for critical infrastructure monitoring, given the high stakes involved.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a distribution grid monitoring project, leveraging Orange's cybersecurity and IoT experts, and incorporating Evolution Platform's capabilities, with a focus on demonstrating isolation and resilience against cyber threats to meet NERC CIP requirements.
Sales	In a conversation with a North American energy utility's head of grid operations, mention Orange's ISO 27001 and SecNumCloud certifications as a foundation for secure network slicing, and ask about their current approach to NERC CIP compliance for grid monitoring.
Strategist / Innovator	Commission a deep-dive brief on how Orange's cybersecurity and IoT expertise can be combined with Evolution Platform to address secure network slicing for distribution grid monitoring, focusing on NERC CIP compliance and isolation mechanisms.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-D8CAA774FDC4	2026-07-01	Next Research	1	Securing 5G network slicing for the smart grid network
SIG-0CB746316F44	2026-05-21	International Journal of ...	1	Industrial Control System Cyber Security Frameworks: Assessment using Defense-in-Depth Criteria
SIG-F633044AF87A	2026-01-29	openalex.org	1	SecuFL-IoT: an adaptive privacy-preserving federated learning framework for anomaly detection i...
SIG-C201BF32AE96	2026-01-07	openalex.org	1	Federated Learning for Secure Industrial Automation and Grid Optimization
SIG-A5AB1DD30A6D	2025-11-10	openalex.org	1	E2E Network Slicing for Enhanced Cybersecurity, Orchestration, Automation and Response in 5G/6G...
SIG-263E4357EB27	2025-11-08	openalex.org	1	Investigating 5G Network Slicing Security Vulnerabilities Using Artificial Intelligence–Driven ...
SIG-EBB2157E5B89	2025-11-01	Acta Scientific Medical S...	1	Resilient AIDriven Network Slicing for DisasterAware 6G Infrastructures
SIG-0065C52738A8	2025-10-29	openalex.org	1	QRoNS: Quantum Resilience over IPsec Tunnels for Network Slicing
SIG-221073AED38A	2025-10-19	2025 IEEE 102nd Vehicular...	1	Designing Secure Network Functions for UE Threat Detection in Network Slicing
SIG-0D07056A550F	2025-10-01	cordis.europa.eu	1	Persistent Awareness system facilitating Customs Checks
SIG-D4CA61FE83AD	2025-08-29	openalex.org	1	AI-powered cyber resilience: a reinforcement learning approach for automated threat hunting in ...
SIG-FFC98154A897	2025-04-14	WSEAS TRANSACTIONS ON COM...	1	Contemporary Network Slicing Security
SIG-2424954F4021	2026-05-24	arxiv.org	3	ASTRO: Adaptive Spatio-Temporal Reinforcement Optimization for GNN Powered Anomaly Detection in ...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: diagram (3 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:15+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:19+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:56:41+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.